

FROZEN SITE-PROBE ACCURACY IS NOT A STANDALONE HARMONIZATION SCORE: AN AUDIT OF EVALUATION PRACTICE FOR MULTI-SITE MRI

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ABSTRACT

Image-level MRI harmonization is often scored by how poorly a frozen site classifier recognizes the acquisition site of harmonized images. We audit this practice on raw ABIDE I T1-weighted MRI (1,112 subjects, 17 sites), using ten outputs from seven harmonization families as test cases. (i) Verdicts depend on how the probe is trained: five probes trained with the benchmark’s recipe gave the same CycleGAN outputs site balanced accuracies of 0.13–0.59; five converged probes agreed (0.44–0.54) but found more site information in eight of ten outputs. (ii) The lowest accuracy, at the level of shuffled-label probes, came from the most destructive output. (iii) Diffusion translation changed intensities least but barely moved them toward the target site. (iv) Low frozen-probe accuracy did not establish removal: probes trained on harmonized outputs still decoded the source site from whole heads, skull-stripped brains and brain intensity histograms. We propose a reporting checklist.

Index Terms— MRI harmonization, evaluation, site classification, domain shift, reproducibility

1. INTRODUCTION

Acquisition differences between sites confound pooled MRI analyses and degrade models on unseen scanners [1, 2]. Image-level harmonization translates scans toward a common appearance. Travelling subjects [3] or paired protocols [4] allow direct checks on few scanners, so large retrospective cohorts are judged indirectly [5]. A common proxy is a *site probe*: a classifier trained to recognize the site on unharmonized images and applied, frozen, to harmonized ones [2, 6]. Lower accuracy is read as removal of site information and is paired with similarity to the input, such as PSNR, and with intensity-distribution distances.

Each proxy measures less than it appears to. A frozen probe reports what one classifier recovers; it can also fail because outputs are unfamiliar, damaged or noisy. Removing

an attribute until a fixed adversary fails does not stop a newly trained classifier from recovering it [7]. Site is also decodable at about 0.9 balanced accuracy from frozen, even randomly initialized, encoders, so the signal lies partly in low-level image statistics [8]. Similarity to the input rewards doing nothing, and a small input-to-output distance says nothing about the target site.

We audit these practices on outputs of seven harmonization families. Unlike feature-level studies showing that site classifiers retrained after ComBat still predict site [9], this is a controlled image-level audit varying probe convergence and training data, separating geometry from intensity, measuring target alignment and including a destructive counterexample. Contributions: (1) a subject-level protocol with paired uncertainty, probe controls and target alignment measured apart from change; (2) evidence that probe verdicts depend on how the probe was trained, that the lowest site accuracy can signal destruction, and that small change does not imply alignment; (3) a test of whether site information stays decodable after harmonization, using probes trained on harmonized images with geometry-only, intensity-only and brain-only controls; and (4) a reporting checklist.¹ We neither propose nor rank harmonizers.

2. DATA AND AUDITED OUTPUTS

Cohort. Raw ABIDE I T1-weighted scans [10] of 1,112 subjects from 17 sites are split by subject, stratified by site, into 890 training, 113 validation and 109 test subjects. Volumes are reoriented, clipped to $[p_1, p_{99}]$ and scaled to $[0, 1]$; one axial slice per subject is chosen by foreground area, cropped to a dilated head mask and resized to 256×256 . NYU, the largest site, is the target; 19 test subjects are from NYU and 90 from 16 source sites. Exact ties made slice choice host-dependent, so slices are frozen and verified for every output. The label is coarser than the acquisition: ABIDE I releases UM, UCLA and Leuven as two sub-samples each, merged by our labels, and 11 of 17 sites hold more than one voxel size

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¹Code, the evaluation protocol with all amendments, and per-subject results: <https://github.com/muqsitamir/HarmonIt>.

or matrix (UCLA includes 10 scans at $1.5 \times 1.5 \times 4$ mm); no per-subject scanner identifier is released.

Audited outputs. Earlier benchmark outputs were evaluated without retraining: NeuroCombat [11, 1] per pixel, fit on the test cohort itself (transductive); histogram matching of foreground quantiles to a pooled 17-site training reference [12]; CycleGAN [13]; StarGAN [14], aggressive and conservative; DLEST [15] after 1,000 and 1,500 steps; conditional diffusion image-to-image translation [16, 17] (strength 0.35, 50 DDIM steps); and adapted volumetric HCLD [18]. Learned methods were trained on the training split. Test subjects were inspected in the earlier benchmark, so this is a retrospective reanalysis; each audit experiment was written into a protocol before its outcome was seen, recording what had been observed.

3. AUDIT PROTOCOL

Change and target alignment. PSNR on a fixed $[0, 1]$ scale and normalized cross-correlation (XCorr) compare output with input on whole images; generators return NYU inputs almost unchanged, so we summarize the 90 source subjects. *Change* is the per-subject Wasserstein-1 distance W between input and output intensities. *Target alignment* is the Wasserstein-1 distance and KL divergence between foreground intensities (defined on the raw slice) and a reference from the 147 NYU training subjects; ΔW_{NYU} is the output’s distance minus the input’s.

Site probes. All probes are single-channel ResNet-18 networks [19] predicting 17 sites, trained with AdamW (learning rate 3×10^{-4} , batch 64, 500 steps) and random affine augmentation, each selected on validation BA from its own training source and evaluated once. We train 44 of them in five families:

Frozen (1): the earlier benchmark’s own checkpoint.

Benchmark recipe (5 seeds): that training repeated on raw volumes. Does the seed change the verdict?

Converged (5 seeds): four times the steps with cosine decay, tested at the final epoch. Do they agree?

Shuffled labels (4): labels permuted between subjects. Is the probe reading images, not labels?

Slice probes (24, plus 6 restricted-input networks and 8 logistic fits): one fixed slice per subject, from raw images or one method’s outputs. Is site still decodable? Restricted variants see only head silhouettes or brain masks (geometry), only foreground intensity histograms (intensity), or slices inside raw-volume HD-BET brain masks [20].

The three slice-probe methods were chosen in advance to span intensity, adversarial and diffusion families; the converged, silhouette, histogram and brain-only probes were added after earlier results but before their own.

Statistics. Source BA averages the 16 source sites, with only 2–15 test subjects each. A uniform guess over the 17

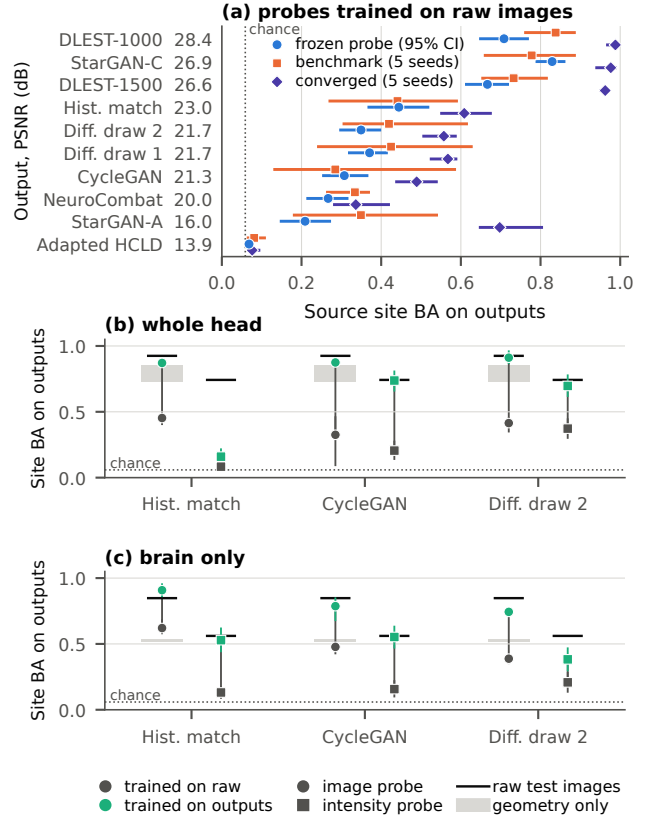


Fig. 1. (a) Source site BA of each output, ordered by PSNR; retrained probes show the mean and range over five seeds. (b, c) Probes trained on raw slices (grey) or on a method’s outputs (aqua), tested on that method’s outputs (diffusion: second draw): image probes (circles, mean and range over 3 seeds) and intensity-histogram probes (squares, 95% interval). Black bar: raw-trained probe on raw test slices. Grey: image probes given only head silhouettes or brain masks.

classes scores $1/17 \approx 0.06$; shuffled-label probes, our empirical null, scored 0.03–0.08. Intervals are 95% percentiles of 2,000 bootstrap replicates resampling subjects within site, paired across methods and probes, and conditional on these sites and checkpoints.

4. RESULTS

Probe validity. On raw source images the frozen probe reached BA 0.95 [0.90, 1.00], benchmark-recipe probes 0.81–0.98 and converged probes 1.00. Probes learn site-related structure, not necessarily scanner appearance: head silhouettes alone gave 0.73–0.86.

Verdicts depend on probe training. Benchmark-recipe probes assigned very different site BA to the same outputs (Table 1, Fig. 1a): CycleGAN 0.13–0.59 and aggressive Star-

Table 1. Source-subject results ($n = 90$). PSNR and XCorr compare output with input; ΔW_{NYU} : change in Wasserstein-1 distance to the NYU reference (negative is closer). Site BA: frozen probe with 95% interval; five benchmark-recipe and five converged probes as mean [min, max]. Uniform chance is 0.06; shuffled-label probes score 0.03–0.08. [†]Fit on the test cohort. [‡]Pooled 17-site reference.

Method	Change and target alignment			Source-site balanced accuracy of probes trained on raw images		
	PSNR	XCorr	ΔW_{NYU}	Frozen probe	Benchmark recipe	Converged recipe
Raw input	–	–	0	0.95	0.90 [0.81, 0.98]	1.00 [1.00, 1.00]
NeuroCombat [†]	20.0	0.940	+0.003	0.27 [0.21, 0.32]	0.33 [0.26, 0.37]	0.34 [0.28, 0.42]
Histogram matching [‡]	23.0	0.982	-0.053	0.44 [0.37, 0.52]	0.44 [0.27, 0.59]	0.61 [0.55, 0.68]
CycleGAN	21.3	0.977	-0.036	0.31 [0.25, 0.37]	0.28 [0.13, 0.59]	0.49 [0.44, 0.54]
StarGAN (aggr.)	16.0	0.896	-0.019	0.21 [0.15, 0.27]	0.35 [0.18, 0.54]	0.70 [0.65, 0.81]
StarGAN (cons.)	26.9	0.991	-0.021	0.83 [0.79, 0.86]	0.78 [0.66, 0.89]	0.98 [0.94, 0.99]
DLEST 1500	26.6	0.993	-0.020	0.67 [0.61, 0.72]	0.73 [0.65, 0.82]	0.96 [0.95, 0.97]
DLEST 1000	28.4	0.993	-0.009	0.71 [0.65, 0.77]	0.84 [0.76, 0.89]	0.99 [0.96, 1.00]
Diffusion img2img	21.7	0.948	-0.001	0.37 [0.32, 0.42]	0.42 [0.24, 0.63]	0.57 [0.52, 0.59]
Adapted HCLD	13.9	0.706	+0.108	0.07 [0.06, 0.08]	0.08 [0.06, 0.11]	0.08 [0.06, 0.10]

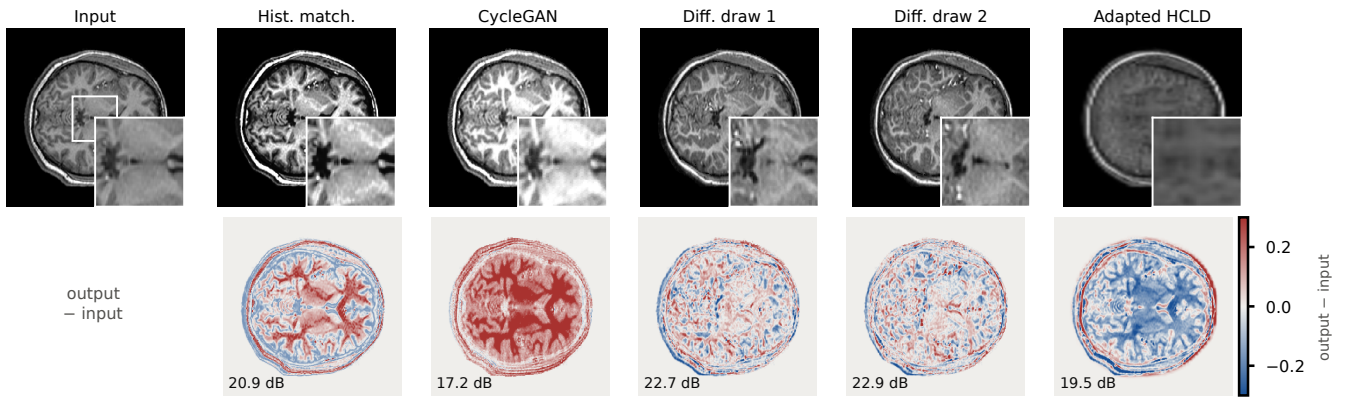


Fig. 2. Source test subject with median PSNR over all outputs; insets are a $2 \times$ zoom of the boxed region. Bottom: output minus input, with this subject’s PSNR. The adapted HCLD output loses anatomy; diffusion adds texture that differs between draws.

GAN 0.18–0.54, with Kendall τ 0.51–0.87 between method orderings. Their validation BA fluctuated between epochs (e.g. 0.88, 0.50, 0.87), and raw accuracy did not reveal it: the probe with raw BA 0.98 gave CycleGAN 0.20. Converged probes (validation BA within 0.035 over the last five epochs) agreed (τ 0.82–0.96; per-output ranges at most 0.16, within the widest single-probe interval of 0.18), but every converged probe exceeded the frozen probe’s interval on all outputs except adapted HCLD and NeuroCombat, e.g. aggressive StarGAN 0.65–0.81 against 0.21 (Table 1). Orderings changed (τ 0.64–0.73 with the frozen probe): it placed aggressive StarGAN second lowest, converged probes above histogram matching. A probe that has not converged overstates harmonization.

The lowest site accuracy came from destruction. The adapted HCLD output reached BA 0.07 [0.06, 0.08] under the frozen probe and 0.06–0.11 under every retrained probe, within the shuffled-label null, with the lowest PSNR (13.9 dB) and XCorr (0.71), visible loss of anatomy (Fig. 2), and intensities moved away from the target ($\Delta W_{\text{NYU}} = +0.108$). It is a counterexample for the metric, not a verdict on the published HCLD method.

Small change is not target alignment. Diffusion translation changed intensities least ($W = 0.008$) yet moved them barely closer to NYU ($\Delta W_{\text{NYU}} = -0.0013$ [−0.0023, −0.0004], 1% of the raw distance) and raised the KL divergence to NYU from 0.30 to 0.83. Its outputs are also a random draw: resampling the same checkpoint changed the image more than harmonization did (per-subject mean absolute foreground difference, median 0.085 against 0.075), at unchanged PSNR.

Is site still decodable? Slice probes trained on raw slices reached BA 0.92 on raw test slices but 0.45, 0.33 and 0.41 on histogram-matched, CycleGAN and diffusion outputs, the drop usually reported as harmonization. Probes trained on each method’s outputs recovered the source site from its test outputs with BA 0.87, 0.87 and 0.91 (Fig. 1b), with disjoint intervals for every seed. These slice probes use the unconverged benchmark recipe (Fig. 1a), so they understate recoverability. Because head geometry identifies sites, we repeated this on brain-only slices: raw-trained probes reached 0.85 on raw brains but 0.62, 0.48 and 0.39 on the outputs, and probes trained on outputs 0.91, 0.79 and 0.74, again disjoint for every seed with best-validation checkpoints (five of nine with

final-epoch ones); brain masks alone gave 0.52–0.54. Intensity histograms remove geometry. On whole heads, a histogram probe trained on outputs recovered little from histogram matching (0.16 [0.11, 0.22]), whose head histograms are matched by construction, but recovered CycleGAN (0.74) and diffusion (0.70) as well as raw slices (0.74). Restricted to the brain it recovered all three: 0.53 [0.44, 0.63], 0.55 [0.47, 0.64] and 0.38 [0.29, 0.47], against 0.56 on raw brains and 0.13–0.21 for the raw-trained probe. Low frozen-probe accuracy therefore does not establish removal: substantial site-label information stays decodable once the probe is adapted to the transformed distribution.

5. DISCUSSION AND RECOMMENDATIONS

Three consequences follow for practice. (i) *One site number is not a verdict.* It moved by up to 0.46 with the probe seed and was lowest for the most destructive output, so read it beside the probe recipe, evidence of convergence and a fidelity measure. (ii) *Train a probe on the outputs.* Probes trained on harmonized images recovered the source site at 0.87–0.91, and 0.74–0.91 within the brain, where raw-trained probes had suggested removal. This shows decodability, not that the recovered signal is scanner-related. (iii) *Measure the target, not the change.* The least-changed output closed 1% of its initial foreground intensity-distribution distance to NYU. Site labels are confounded: head and brain shape predict them, so image-probe accuracy cannot certify scanner-effect removal.

We recommend reporting (1) several converged probes with their recipe, including output-trained and channel-restricted probes with geometry controls; (2) each probe’s raw accuracy and a shuffled-label null; (3) target alignment separately from change, excluding target-site subjects; and (4) paired uncertainty over subjects and generator draws.

Limitations. We use one cohort, one target site and one axial slice per subject, so volumetric effects are not assessed. Without anatomical ground truth, PSNR and XCorr measure agreement with the input, not correctness. Brain masks leave brain shape, slice level and orientation, and site is confounded with demographics and with acquisition geometry that varies within a site; our controls bound these channels but cannot separate scanner effects from biological differences. Generators were not retuned, and one baseline is transductive.

6. COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using human subject data made available in open access by ABIDE I [10], whose sites shared the fully anonymized data with local ethics approval or an explicit waiver. Ethical approval was not required as confirmed by the license attached with the open access data.

7. ACKNOWLEDGMENTS

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